

SURFACE PREPARATION

The Foundation of Every Flawless Finish

Skip the prep. Ruin the part.

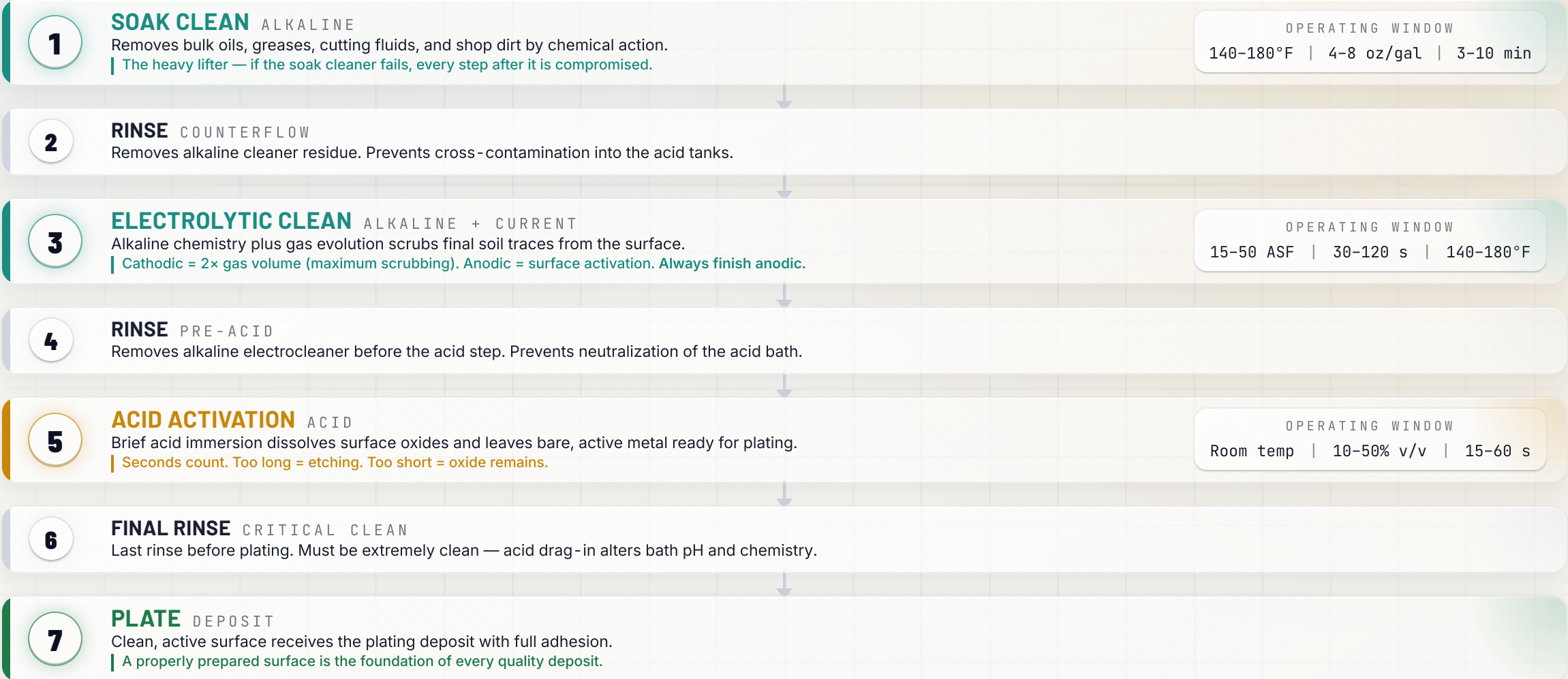
THE 80% RULE

80%+

of plating defects originate in surface preparation — not the plating bath.

THE PRE-TREATMENT SEQUENCE

Every plating line. Every substrate. This sequence — or a variation of it — runs before every deposit.



THE WATER BREAK TEST

A 30-second field check that catches failed prep before it costs you a rework.

• CLEAN



Water sheets uniformly across the surface

✓ PASS – PROCEED TO PLATE

• CONTAMINATED



Water beads up — oil or residue present

✗ FAIL – RETURN TO CLEANING

► IF THE WATER BEADS, DON'T PLATE. ◀

SUBSTRATE-SPECIFIC SEQUENCES

SUBSTRATE	SEQUENCE VARIATION	KEY CONSIDERATION
Steel / Iron	Standard sequence	H ₂ embrittlement risk above 39 HRC — minimize cathodic; bake within 4 hrs
Aluminum	Caustic etch + desmut	Al ₂ O ₃ oxide requires alkaline removal; desmut removes smut
Zinc Die-Cast	Anodic EC only + mild H ₂ SO ₄ + Cu strike	Porous — extended soak clean essential; cathodic EC deposits smuts
Stainless Steel	Standard + Wood's nickel strike	Passive Cr oxide requires Wood's (NiCl ₂ , pH <1.0, 50–250 ASF)
Copper / Brass	Standard + optional bright dip	Tarnishes rapidly — minimize acid-to-plate transfer time

*All variations begin with soak clean and end with plate. Consult your process supplier for product-specific recommendations.

GOVERNING STANDARDS

ASTM B322	Standard Guide for Cleaning Metals Prior to Electroplating
ASTM B850	Post-Coating Hydrogen Embrittlement Relief
AMS 2759/9	Hydrogen Embrittlement Relief for Steel Parts (Aerospace)

WHEN PREP FAILS

BLISTERING Residual oil or oxide under deposit	SKIP PLATING Passive oxide not removed	PITTING H ₂ trapped by organic contamination
PEELING Contamination between substrate and plate	ROUGHNESS Particulate from cleaning tanks	STAINING Alkaline or acid drag-out

⚠ WATCH YOUR ELECTROCLEANER

Chloride contamination above **10 g/L** in the electrocleaner causes initial corrosion embedded in the substrate before plating. When plated over, this invisible damage leads to premature salt-spray failure.

| Titrate for chlorides periodically. Run a reverse-current Hull cell if etching is suspected.